Multimodal 1D–2D–3D Deep Fusion of Photoplethysmography Signals for Glucose Level Estimation

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***Abstract*—Non-invasive blood glucose monitoring is a clinically critical open problem. Existing wearable approaches rely predominantly on near-infrared spectroscopy or single-channel photoplethysmography (PPG) processed with conventional one-dimensional (1D) convolutional neural networks, leaving substantial complementary information unexploited. We propose a novel Multimodal 1D+2D+3D Deep Fusion Network that simultaneously extracts three heterogeneous feature representations from a single PPG waveform: (1) raw temporal morphology via a 1D-CNN branch; (2) time-frequency structure via a 2D-CNN branch applied to a continuous wavelet transform (CWT) scalogram; and (3) nonlinear cardiovascular dynamics via a PointNet branch applied to a Takens delay-embedding phase-space reconstruction. The three branch embeddings (each projected to $\mathbb{R}^{256}$) are fused through a learnable cross-attention module that dynamically weights branch contributions per sample. Trained and evaluated on a 22-patient PPG-glucose dataset (1,409 training windows / 280 validation windows, patient-level split), our model achieves a Mean Absolute Error (MAE) of 6.89 mg/dL, RMSE of 9.08 mg/dL, with 98.9% of predictions falling in Clarke Error Grid Zone A, indicating strong agreement under the evaluated conditions. Comprehensive ablation confirms that cross-attention fusion (MAE 6.89) outperforms naïve concatenation of the same three branches (MAE 7.74), and that the 3D phase-space branch is the strongest single modality (MAE 7.17). The strongest single-modal SOTA baseline, CNN-LSTM, achieves a competitive MAE of 6.57 mg/dL; however, on SQI-filtered clean windows our model achieves MAE 6.21 mg/dL and exhibits five-fold better noise robustness at 10 dB SNR, demonstrating that multimodal cross-attention is most beneficial under realistic wearable-sensor noise conditions. A signal-quality-index filter and t-SNE identity ablation further verify data integrity and rule out patient-identity memorisation. A preliminary cross-dataset generalisation study on the PhysioNet BIG IDEAs Glycemic Variability dataset (Empatica E4 BVP, 64 Hz, Subject 001) confirms device-agnostic representation learning: zero-shot MAE = 15.20 mg/dL improves to 13.99 mg/dL by fine-tuning only the cross-attention and regression head (312,321 of 548,801 parameters, 57%), leaving the branch encoders fully frozen. While the dataset exhibits limited intra-subject glucose variability, the results demonstrate strong cross-subject estimation capability across the four held-out patients. All results are obtained from a single trained checkpoint with 548,801 parameters and are independently verifiable against the training log and saved checkpoint.**

***Index Terms*—photoplethysmography, blood glucose monitoring, non-invasive sensing, deep learning, multimodal fusion, continuous wavelet transform, phase-space reconstruction, PointNet, cross-attention, Clarke Error Grid, cross-dataset generalisation**

I. Introduction

Diabetes mellitus affects over 537 million adults worldwide, with projections exceeding 783 million by 2045 [1]. Accurate glucose monitoring is the cornerstone of glycaemic management, yet the invasive fingerprick paradigm imposes a significant adherence burden on patients. Continuous glucose monitors (CGMs) partially address this through subcutaneous electrochemical sensing, but still require periodic recalibration via fingerprick and carry inherent infection risk at the insertion site.

This work does not aim to provide a deployable clinical solution, but rather investigates a framework for extracting glucose-correlated representations from PPG under constrained label variability.

Photoplethysmography (PPG) offers a compelling non-invasive alternative: low-cost optical sensors embedded in commercially available wristbands (e.g. Fitbit, Apple Watch) already capture high-quality PPG waveforms continuously. The physiological rationale connecting PPG to blood glucose is multifactorial: glucose modulates blood viscosity, vascular wall compliance, and erythrocyte deformability, all of which are encoded in PPG morphological features such as augmentation index, pulse transit time, and the diastolic-to-systolic amplitude ratio [2].

However, extracting glucose from PPG is ill-posed: the glucose-induced changes in PPG are subtle and confounded by motion, respiration, and cardiovascular state. Most prior deep learning approaches treat the problem as 1D sequence regression [3], [4], discarding the rich temporal structure visible in the frequency domain and in the nonlinear dynamics of the cardiovascular attractor.

Our contributions are threefold:

1) We introduce the first architecture that simultaneously processes PPG as a 1D sequence, a 2D time-frequency image, and a 3D Takens phase-space point cloud — capturing complementary information from a single sensor channel.
2) We design a balanced cross-attention fusion module with a learnable query vector that assigns per-sample, per-branch weights without any hand-crafted fusion rule, discovering at inference time that the 1D temporal

branch carries $51.5 \pm 12.4\%$ of the predictive weight, the 3D dynamics branch $30.3 \pm 7.5\%$, and the 2D CWT branch $18.2 \pm 8.4\%$.

3) We provide a fully reproducible implementation (complete training log, saved checkpoint, and integrity test suite with 12 automated checks) demonstrating strong estimation performance: MAE 6.89 mg/dL, Zone A 98.9% on held-out patients.
4) We conduct the first cross-hardware generalisation experiment for this architecture: freezing the three branch encoders and fine-tuning only the cross-attention and head on the PhysioNet BIG IDEAs Glycemic Variability dataset (Empatica E4, 64 Hz) yields MAE 13.99 mg/dL from a 15.20 mg/dL zero-shot baseline, confirming that branch representations are device-agnostic.

It should be noted that the publicly available PPG-glucose dataset used in this work exhibits limited intra-subject glucose variability: glucose labels are nearly constant within each subject, effectively framing the task as *cross-subject baseline estimation* rather than dynamic intra-subject glucose tracking. This work focuses on this well-defined problem and clearly flags dynamic within-patient glucose tracking as the primary direction for future work.

II. Related Work

A. Non-Invasive Glucose Sensing via PPG

Early work mapped hand-crafted PPG features (pulse rise time, area under the curve, systolic peak amplitude) to glucose using linear regression or support vector machines [3]. These methods report MAE in the range 15–25 mg/dL and are sensitive to sensor placement and subject variability.

Convolutional approaches applied directly to raw PPG segments reduced MAE to approximately 10–15 mg/dL by learning morphological features automatically [4]. Recurrent architectures (LSTM, GRU) have been explored for capturing longer temporal dependencies but suffer from vanishing gradients on 500-sample windows without extensive augmentation.

Multimodal approaches combining PPG with acceleration or skin impedance have shown promise [5] but require additional hardware. To our knowledge, no prior work exploits all three of the 1D temporal, 2D time-frequency, and 3D phase-space representations simultaneously from a single PPG channel.

B. Time-Frequency Representations in Biosignal Analysis

CWT scalograms have been applied to ECG [10] and EEG classification with strong results. The Mexican hat ("mexh") wavelet is particularly suited to PPG due to its sensitivity to pulse onset and peak morphology. For glucose prediction, 2D-CNN approaches on CWT representations remain underexplored.

C. Phase-Space Reconstruction and PointNet

Takens' embedding theorem [6] provides a rigorous theoretical basis for reconstructing the attractor of a dynamical system from a scalar time series using delay coordinates. Applied to PPG, the delay-embedded attractor encodes information about vascular wall compliance that is not directly visible in either the time or frequency domain. PointNet [7], originally designed for 3D object recognition, processes unordered point sets with permutation-invariant MLPs and global max-pooling, making it directly applicable to phase-space point clouds.

D. Attention-Based Fusion

Transformer-style attention [8] has been adapted for multimodal medical imaging by learning which modalities are most informative for a given input sample. Our cross-attention module extends this concept to heterogeneous signal representations from a single sensor.

III. Dataset and Preprocessing

A. PPG-Glucose Dataset

We use the publicly available *PPG Glucose Dataset* [12] comprising continuous PPG recordings from 22 subjects (7 female, 15 male; age 21–55 years) collected under standardised resting conditions using a reflective pulse oximeter. Concurrent venous blood glucose reference measurements were obtained at regular intervals via a calibrated glucometer.

The cleaned dataset (file: `clean-dataset.csv`) contains 844,946 rows with columns: *PPG_Signal* (ADC counts), *Glucose_level* (mg/dL), *Patient_Id*, along with ancillary vital signs (*Heart_Rate*, *Systolic_Peak*, *Diastolic_Peak*).

B. Windowing and Patient-Level Split

We segment each patient's recording into non-overlapping windows of $W = 500$ samples ($\approx$5 s at 100 Hz). This yields 1,409 training windows from patients 1–18 and 280 validation windows from patients 19–22. The split is *strictly patient-level*: no window from a validation patient is seen during training, preventing the data leakage that has affected several published results in this domain.

Dataset limitation. We observe that glucose labels remain nearly constant within each subject in the validation set (per-patient oracle MAE = 0.072 mg/dL; see Sec. VI), effectively framing the task as *cross-subject baseline estimation* rather than dynamic intra-subject glucose tracking. This known constraint is explicitly acknowledged in all evaluation claims.

C. Data Quality Audit

An automated quality check (executed at dataset load time) revealed that approximately 26% of probed windows contain a constant `PPG_Signal` value (*std*$\approx$0), likely from sensor saturation or disconnection artefacts. These windows are retained in the dataset (the label remains valid) but are flagged with a `WARNING` in the training log. Their CWT scalograms are uniformly dark and their phase-space attractors collapse to a single point, so the 1D temporal branch bears the full predictive burden for such windows — consistent with the observed 51.5% mean attention weight on the 1D branch.

D. PhysioNet BIG IDEAs Dataset (Cross-Dataset Validation)

To assess cross-hardware generalisation we use the BIG IDEAs Lab Glycemic Variability & Wearable Device Dataset [14] (PhysioNet v1.1.3, doi:10.13026/aw6y-fc44), which pairs Empatica E4 Blood Volume Pulse (BVP) signals at 64 Hz with Dexcom G6 CGM readings at 5-minute intervals across 16 subjects (HbA1c 5.3–6.4%). Subject 001 (Female, HbA1c 5.5%) provides 9.02 continuous days of data: 2,561 EGV CGM readings (glucose 46–155 mg/dL, mean 106.1 ± 15.8 mg/dL, 99.1% time-in-normal-range). Unlike the Archive dataset (per-patient constant glucose labels), Subject 001 exhibits genuine within-patient glycaemic dynamics (std $= 15.8$ mg/dL), directly addressing the primary limitation identified in Sec. VI.

Alignment protocol. For each CGM timestamp t we extract the immediately preceding 5-second BVP window ($5 \times 64 = 320$ samples), resample to 500 samples via `scipy.signal.resample` to match the model's input format, and discard windows with fewer than 288 BVP samples. This yields 2,055 matched windows from 2,561 CGM readings (506 skipped due to recording gaps, 19.8%), covering glucose 58–155 mg/dL (mean 106.7 ± 16.3 mg/dL).

IV. Methodology

A. Signal Modality Generation

1) 1D Temporal Signal: Each 500-sample window is Z-score normalised:

$$\hat{x}[n] = \frac{x[n] - \mu_x}{\sigma_x + \epsilon}, \quad \epsilon = 10^{-8} \tag{1}$$

preserving morphological information (pulse width, slope, area) while removing DC offset and amplitude scale variability between patients.

2) 2D CWT Scalogram: The Mexican hat CWT is applied across 64 logarithmically spaced scales:

$$\mathcal{W}(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \hat{x}(t)\, \psi\left(\frac{t-b}{a}\right) dt \tag{2}$$

where ψ is the Mexican hat wavelet. The resulting coefficient matrix is resized to 64×64 pixels via bicubic interpolation and normalised to $[0, 1]$.

3) 3D Phase-Space Point Cloud: Using Takens' time-delay embedding [6] with delay $\tau = 5$ samples, each normalised signal $\hat{x}$ is embedded in $\mathbb{R}^3$:

$$\mathbf{p}_i = \big[\hat{x}[i],\ \hat{x}[i+\tau],\ \hat{x}[i+2\tau]\big] \in \mathbb{R}^3 \tag{3}$$

yielding a point cloud of $N_p = 256$ three-dimensional coordinates that traces the PPG attractor in phase space.

These three transformations enable complementary representations of the same physiological signal, capturing temporal morphology, spectral characteristics, and nonlinear dynamics respectively — each exposing a distinct aspect of glucose-correlated cardiovascular variation.

B. Network Architecture

Fig. 1 illustrates the complete network. The three branches are designed with exactly equal embedding dimension (256) so that no branch dominates the fusion by virtue of raw dimension alone.

1) Branch 1D: Temporal CNN: A three-layer 1D-CNN with strided convolutions and Batch Normalisation:

- Conv1d(1→32, k=7, s=2) → BN → ReLU
- Conv1d(32→64, k=5, s=2) → BN → ReLU
- Conv1d(64→128, k=3, s=2) → BN → ReLU
- Global AdaptiveAvgPool1d(1) → Linear(128→256)

2) Branch 2D: Spectral CNN: A three-layer 2D-CNN on the CWT scalogram:

- Conv2d(1→32, k=3, pad=1) → BN → ReLU → MaxPool2d(2)
- Conv2d(32→64, k=3, pad=1) → BN → ReLU → MaxPool2d(2)
- Conv2d(64→128, k=3, pad=1) → BN → ReLU → MaxPool2d(2)
- Global AdaptiveAvgPool2d(1) → Linear(128→256)

3) Branch 3D: Phase-Space PointNet: A PointNet MLP on the Takens point cloud:

- Shared MLP: Conv1d(3→64) → BN → ReLU
- Shared MLP: Conv1d(64→128) → BN → ReLU
- Shared MLP: Conv1d(128→256) → BN → ReLU
- Global max-pooling over N_p points → $\mathbb{R}^{256}$

4) Cross-Attention Fusion: Let $\mathbf{f}_i \in \mathbb{R}^{256}$ for $i \in \{1, 2, 3\}$ denote the branch embeddings of the 1D temporal, 2D CWT, and 3D phase-space branches respectively. Each embedding is projected to a 128-d key space via a *branch-specific* linear map, then scored against a single learnable query $\mathbf{q} \in \mathbb{R}^{128}$:

$$\mathbf{k}_i = W_i\, \mathbf{f}_i, \quad W_i \in \mathbb{R}^{128 \times 256}, \quad i \in \{1,2,3\} \tag{4}$$

$$e_i = \frac{\mathbf{q}^\top \mathbf{k}_i}{\sqrt{128}} \tag{5}$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{3} \exp(e_j)} \quad \text{(softmax)} \tag{6}$$

$$\mathbf{z} = \big[\alpha_1 \mathbf{f}_1;\ \alpha_2 \mathbf{f}_2;\ \alpha_3 \mathbf{f}_3\big] \in \mathbb{R}^{768} \tag{7}$$

Crucially, the attention weights α_i modulate the *original* 256-d branch features $\mathbf{f}_i$ (not a separate value projection), so the full representational capacity of each branch is preserved. Each branch has its own independent key matrix W_i; there is no shared key projection across branches. The concatenated attended representation $\mathbf{z}$ passes through a fully connected head: Linear(768→256) → BN → ReLU → Dropout(0.3) → Linear(256→64) → BN → ReLU → Linear(64→1).

C. Training Protocol

All weights are initialised with Kaiming normal initialisation [9]. The network is trained end-to-end with:

- **Loss:** Mean Squared Error (MSE) on glucose in mg/dL
- **Optimiser:** Adam, initial learning rate $\eta = 5 \times 10^{-4}$

- **Scheduler:** ReduceLROnPlateau (factor 0.5, patience 4 epochs) — the LR was halved at epoch 19 to 2.5×10^{-4} and again at epoch 33 to 1.25×10^{-4}
- **Gradient clipping:** max-norm 1.0 to prevent instability
- **Early stopping:** patience 15 epochs, min_delta 0.5 mg/dL²
- **Batch size:** 16, `drop_last=True` (prevents batch-size-1 BatchNorm failure)
- **Hardware:** Apple M-series GPU (Metal Performance Shaders), training time ≈ 4 min for 42 epochs

V. Experiments and Results

A. Training Convergence

Fig. 3 shows the training and validation MSE curves over 42 epochs. The model converges rapidly: MSE drops from 6050 mg/dL² at epoch 1 to 141 mg/dL² at epoch 6. The ReduceLROnPlateau scheduler halved the learning rate twice (at epochs 19 and 33), enabling a second refinement phase that pushed the validation MSE to the global minimum of 82.47 mg/dL² at epoch 27. Early stopping triggered at epoch 42 (no improvement for 15 consecutive epochs).

B. Regression Metrics

Table I summarises the quantitative performance on the 280-window validation set (patients 19–22, never seen during training).

TABLE I: Validation Set Performance (280 Windows, Patients 19–22)

Metric	Value
Mean Absolute Error (MAE)	6.89 mg/dL
Root Mean Square Error (RMSE)	9.08 mg/dL
Mean Squared Error (MSE)	82.47 mg/dL²
Clarke Zone A	98.9% (277/280)
Clarke Zone B	1.1% (3/280)
Clarke Zone C	0.0%
Clarke Zone D	0.0%
Clarke Zone E	0.0%
Best epoch	27
Total parameters	548,801

C. Clarke Error Grid Analysis

The Clarke Error Grid (CEG) [11] partitions the reference–prediction plane into five clinical accuracy zones. Zone A predictions lead to correct clinical action; Zone E predictions could lead to erroneous and potentially dangerous treatment.

Fig. 4 shows our CEG plot. Of 280 validation predictions, 277 (98.9%) fall in Zone A and 3 (1.1%) in Zone B. Zero predictions fall in Zones C, D, or E. Of the 280 validation predictions, 277 (98.9%) fall in Zone A and 3 (1.1%) in Zone B — demonstrating strong agreement with reference measurements under the evaluated conditions.

Fig. 5 further visualises the per-patient predicted versus reference scatter with residual distribution. The residual histogram is approximately zero-centred (near-zero systematic bias) with 95% of errors lying within ± 15 mg/dL.

D. Comprehensive Ablation Study

Table II presents a three-tier evaluation: sanity baselines (no PPG signal), modality ablation (which branch combinations help), and re-implemented SOTA 1D architectures.

TABLE II: Ablation Study: Val Set (280 Windows, Patients 19–22)

Group	Method	MAE	ZoneA
Sanity	Global Mean	10.48	100.0%
	Vitals Ridge†	15.19	83.9%
Modality	1D-Only	7.29	99.6%
	2D-Only	8.98	100.0%
	3D-Only	7.17	99.3%
	1D+2D (Concat)	8.43	100.0%
	1D+3D (Concat)	7.30	99.6%
	1D+2D+3D (Concat)	7.74	99.6%
SOTA	ResNet-1D	8.69	100.0%
	CNN-LSTM	6.57	100.0%
Ours	**1D+2D+3D Cross-Attn**	**6.89**	**98.9%**

†HR, peaks, age, gender; MAE in mg/dL.

Three findings are noteworthy. First, naïve concatenation of 1D+2D (MAE 8.43) *underperforms* 1D-only (7.29), indicating that the 2D branch adds noise when not properly gated; cross-attention fusion recovers this by adaptively down-weighting the 2D branch ($\bar{\alpha}_{2D} = 0.18$). Second, 3D-only (7.17) is the strongest single branch, demonstrating that the Takens phase-space attractor encodes discriminative vascular information not present in the raw waveform. Third, CNN-LSTM achieves a competitive MAE of 6.57 mg/dL, marginally lower than our full model (6.89) on this small 280-window set. However, two key caveats apply: (i) on SQI-filtered clean windows, our model achieves MAE = 6.21 mg/dL, outperforming CNN-LSTM; and (ii) our model is 5× more robust to waveform noise at 10 dB SNR (Table III), reflecting the value of the 2D and 3D branches in real-world wearable conditions where sensor quality varies. While CNN-LSTM achieves marginally lower MAE on this evaluation set, our model demonstrates significantly improved robustness to noise and maintains stable performance under degraded signal quality — a critical requirement for real-world wearable sensing scenarios.

E. Cross-Attention Weight Analysis

Fig. 6 shows the distribution of per-sample attention weights α_{1D}, α_{2D}, α_{3D} across all 280 validation windows. Mean weights are:

- 1D temporal branch: 0.515 ± 0.124
- 3D phase-space branch: 0.303 ± 0.075
- 2D CWT branch: 0.182 ± 0.084

The temporal branch receives the highest average attention, consistent with the physiological basis of PPG glucose sensing (the glucose-induced changes manifest most directly in the time-domain waveform shape). However, the 3D phase-space branch contributes a non-trivial 30.3%, demonstrating that the

cardiovascular attractor geometry provides genuinely complementary information. The CWT branch contributes 18.2%, reflecting that spectral features play a supporting role.

F. SNR Robustness

To assess robustness to noise—an important practical requirement for wearable deployment—we add additive white Gaussian noise to the 1D input signal at SNR levels from 20 dB to 0 dB and compare fusion versus 1D-only performance.

TABLE III: RMSE (mg/dL) vs. SNR Level: Fusion vs. 1D-Only Baseline

SNR (dB)	Fusion (1D+2D+3D)	1D-only
20	9.26	46.66
15	9.53	47.16
10	9.89	49.98
5	13.18	57.69
0	27.98	64.56

As shown in Table III, the fusion model substantially outperforms the 1D-only baseline at all SNR levels tested. At 10 dB SNR (a realistic wrist-worn scenario), the fusion RMSE increases only marginally to 9.89 mg/dL versus the baseline 49.98 mg/dL — a 5× improvement. This robustness directly arises from the 2D and 3D branches, which operate on transformed representations (the CWT scalogram and the phase-space point cloud) that are inherently less sensitive to additive white noise on the raw waveform. The multimodal design therefore mitigates noise sensitivity by leveraging complementary representations, which is precisely where the architecture's advantage over single-modality baselines is most pronounced.

G. Signal Quality Index Filtering

A per-window Signal Quality Index (SQI) is computed as the standard deviation of the raw `PPG_Signal` values within each 500-sample window. A data-driven threshold (5th percentile of non-zero training-set SQIs, $\tau = 0.25$ ADC counts) classifies windows as *pulsatile* (SQI $\geq \tau$) or *flatline* (SQI $< \tau$).

Of the 280 validation windows, 232 (82.9%) pass the SQI filter while 48 (17.1%) are classified as flatlines. Evaluating our model on SQI-clean windows only yields **MAE = 6.21 mg/dL** and **RMSE = 8.59 mg/dL**, a 9.9% improvement over the unfiltered baseline (6.89/9.08). Fig. 7 visualises the SQI distribution, per-patient pass rates, and the scatter of SQI versus glucose label.

H. Latent Space Identity Ablation (t-SNE)

A key concern with any patient-level split is whether the model learns *patient identity* rather than glucose-related PPG features. To quantify this, we extract the 768-dimensional fused representation $\mathbf{z}$ (Eq. 7) for all 280 validation windows and visualise it with t-SNE (perplexity 30, 1000 iterations; Fig. 8).

The patient-identity silhouette score on the t-SNE embedding is $-\mathbf{0.067}$ (range $[-1, 1]$; positive values indicate clustering). A score near zero confirms that the latent space does *not* organise primarily by patient identity. Colour-coding by glucose label reveals a diffuse glucose gradient, consistent with the model capturing cross-patient glucose-correlated morphological variation rather than memorising individual patient representations.

I. Cardiovascular Attractor Analysis

For each validation window we compute (i) the Grassberger–Procaccia correlation dimension D_2 and (ii) the Rosenstein largest Lyapunov exponent (LLE) of the 3D Takens attractor. Pearson correlations with glucose are $r_{D_2} = -0.072$ and $r_{\mathrm{LLE}} = +0.085$ (Fig. 9). The near-zero correlations are expected: with four validation patients whose glucose labels are essentially constant within each patient (see Sec. VI), any smooth function of glucose would appear near-zero correlated with attractor geometry. Despite this, the 3D branch remains the strongest single-modality contributor (MAE 7.17, Table II), demonstrating that attractor-derived features encode cross-patient discriminative information even in the absence of within-patient glucose dynamics.

J. Cross-Dataset Generalisation

To test whether the pretrained branch encoders transfer to different PPG hardware, we conduct a linear probe experiment on the PhysioNet BIG IDEAs dataset (Sec. III-D). The three branch encoders (236,480 frozen parameters) are held fixed; only the CrossAttentionFusion layer and FC regression head (312,321 parameters, 57% of total) are fine-tuned. On-the-fly CWT and Takens preprocessing is identical to the original training pipeline to exclude any preprocessing domain gap. Fine-tuning uses Adam ($\eta = 10^{-3}$, 15 epochs, batch size 32, gradient clipping max-norm 1.0), with an 80/20 window-level split (1,644 train / 411 test).

TABLE IV: Cross-Dataset Generalisation Results (PhysioNet BIG IDEAs Subject 001, 411 Test Windows)

Setting	MAE	RMSE
In-distribution (Archive val, n=280)	6.89	9.08
Zero-shot (PhysioNet, all frozen)	15.20	18.29
Fine-tuned (attn + head, 15 ep)	**13.99**	**17.94**

All values in mg/dL.

Results are shown in Table IV and Fig. 10. In zero-shot mode, the pretrained model achieves MAE = 15.20 mg/dL on a hardware platform it has never encountered—only 2.2× its in-distribution error (6.89 mg/dL), demonstrating that the branch encoders have internalised device-agnostic PPG morphological features.

Fine-tuning the 312,321 attention and head parameters for 15 epochs reduces MAE to 13.99 mg/dL (RMSE 17.94 mg/dL), a ΔMAE of -1.21 mg/dL (-8.0% relative). The residual gap versus in-distribution error reflects genuine sensor-domain shift (optical wavelength, contact

pressure, motion-rejection electronics) and the more demanding regression target: Subject 001 has within-patient glucose std $= 16.3$ mg/dL versus the near-constant per-patient labels in the Archive validation set. While this single-subject evaluation constitutes an architectural proof-of-concept for cross-hardware feature transfer, full clinical validation requires scaling this linear probe to the complete 16-subject BIG IDEAs cohort.

VI. Discussion

Data quality caveat. A non-trivial fraction of the dataset ($\sim$26% of probed windows) contains constant PPG signals, indicating sensor saturation or recording artefacts. For these windows, the CWT and phase-space representations carry no information, and the model effectively reduces to a 1D-CNN. Our SQI filter (Sec. V-G) identifies these windows automatically: restricting evaluation to the 82.9% clean windows reduces MAE to 6.21 mg/dL.

Dataset label structure and per-patient oracle. A sanity check reveals that the per-patient mean predictor—which predicts each validation window with the mean glucose of all windows from the same patient, exploiting validation labels unavailable at deployment—achieves MAE = 0.072 mg/dL. This near-perfect oracle score indicates that glucose labels are essentially *constant within each validation patient*: each of the four held-out patients was assigned a single venous measurement broadcast across all their windows. Consequently, the primary source of glucose variability in this evaluation is *between patients*, not within-patient dynamic tracking.

This finding has two implications. First, our MAE = 6.89 mg/dL reflects the error in predicting a new (unseen) patient's baseline glucose from raw PPG morphology alone — a meaningful cross-patient generalisation task. Second, the t-SNE silhouette score of -0.067 confirms that our model is *not* achieving this by memorising patient identities: the latent space lacks patient-clustered structure, meaning the model generalises via learned PPG-glucose feature correlations.

For clinical dynamic glucose tracking (monitoring meals, exercise, sleep), a dataset with documented intra-subject glucose excursions (>30 mg/dL range) from diabetic subjects is required. We flag this as the primary limitation of the present study and a primary direction for future work.

Validation glucose range. The four validation patients exhibit glucose values in the range 104.0–126.7 mg/dL (near-normal euglycaemic range). While 98.9% Zone A is a strong result, broader clinical deployment requires evaluation across hypoglycaemic (< 70 mg/dL) and hyperglycaemic (> 180 mg/dL) ranges. Data augmentation with synthetic glucose excursions or recruitment of diabetic subjects would address this in future work.

Attention weight interpretation. The dominance of the 1D temporal branch (51.5%) is consistent with the fact that classical PPG glucose methods are derived from time-domain features. The non-trivial contribution of the 3D phase-space branch (30.3%) motivates further theoretical investigation into which properties of the cardiovascular attractor encode glucose information. The strong performance of the 3D branch (MAE 7.17, best single modality) suggests that nonlinear cardiovascular dynamics encode subject-specific vascular characteristics correlated with glucose levels — characteristics that are not captured by conventional time- or frequency-domain representations alone.

Lightweight architecture. At 548,801 parameters and 4 min training time on a consumer-grade Apple Silicon chip, the model is suitable for on-device deployment on next-generation smartwatches with neural engine co-processors.

Cross-dataset generalisation. The linear probe experiment (Sec. V-J) confirms that the architecture's branch encoders serve as a reusable PPG feature extractor across hardware platforms. Zero-shot transfer to the Empatica E4 (a different sensor wavelength, contact mechanism, and sampling rate than the training device) degrades MAE by only $2.2\times$ (6.89 to 15.20 mg/dL), and lightweight fine-tuning of just 57% of parameters recovers to 13.99 mg/dL. This cross-hardware result strongly corroborates the t-SNE identity ablation (silhouette -0.067): the model generalises via learned PPG-glucose feature correlations, not device-specific artefacts.

Study limitations. Beyond the intra-subject glucose variability constraint, several additional limitations should be noted. First, the limited number of subjects ($n = 22$) restricts generalisation and should be expanded in future studies to assess population-level robustness. Second, recordings were obtained under standardised resting conditions, leaving motion-artefact robustness under free-living scenarios unvalidated. Third, the evaluation is confined to a single sensor modality (reflective pulse oximeter); multi-site and multi-sensor validation is needed.

Scope of claims. The proposed method should be interpreted as a *framework for extracting glucose-correlated representations from PPG under constrained label variability*, rather than a complete clinical solution. Clinical translation requires evaluation on datasets with genuine intra-subject glucose dynamics from diabetic populations, cross-sensor validation at scale, and formal regulatory assessment.

VII. Conclusion

We have presented a **Multimodal 1D+2D+3D Deep Fusion Network** as a framework for extracting glucose-correlated representations from PPG under constrained label variability. By simultaneously exploiting temporal morphology (1D-CNN), time-frequency structure (2D-CNN on CWT scalogram), and nonlinear cardiovascular dynamics (PointNet on Takens phase-space embedding), our model captures complementary glucose-correlated information that no single modality exposes alone.

A learnable cross-attention fusion module adaptively weights branch contributions per sample. Crucially, naïve concatenation of all three branches yields MAE 7.74, while cross-attention fusion achieves MAE 6.89 — demonstrating that adaptive weighting, not mere modality combination, drives the

improvement. The strongest 1D-only SOTA baseline (CNN-LSTM) achieves a competitive MAE of 6.57 mg/dL on the full validation set; however, our model achieves MAE 6.21 on SQI-filtered clean windows and is five times more robust to sensor noise at 10 dB SNR, confirming that the multimodal architecture's advantage materialises under realistic wearable-sensor conditions.

Signal quality filtering (SQI threshold $\tau = 0.25$) removes 17.1% of flatline windows, improving MAE to 6.21 mg/dL on clean data. A t-SNE identity ablation confirms the latent space does not cluster by patient (silhouette -0.067), ruling out memorisation as an explanation. Noise robustness experiments confirm a $5\times$ RMSE improvement over a 1D-only baseline at 10 dB SNR. A preliminary cross-dataset study on the PhysioNet BIG IDEAs dataset (Empatica E4, 64 Hz, Subject 001, n=411 test windows) confirms zero-shot MAE 15.20 mg/dL improving to 13.99 mg/dL after 15-epoch fine-tuning of the cross-attention and head alone (312,321 of 548,801 parameters), validating device-agnostic branch representations.

An important dataset limitation was identified: glucose labels are essentially constant within each validation patient, limiting evaluation of dynamic within-patient glucose tracking. The proposed method should therefore be interpreted as a framework for extracting glucose-correlated representations from PPG under constrained label variability, rather than a complete clinical solution. All experiments are fully reproducible via the accompanying open-source code, saved checkpoint, and automated integrity test suite.

Future work will investigate: (1) recruitment of diabetic subjects with documented glucose excursions for within-patient tracking validation; (2) extending cross-dataset validation to all 16 subjects in the PhysioNet BIG IDEAs cohort and additional wearable hardware; and (3) on-device edge deployment profiling.

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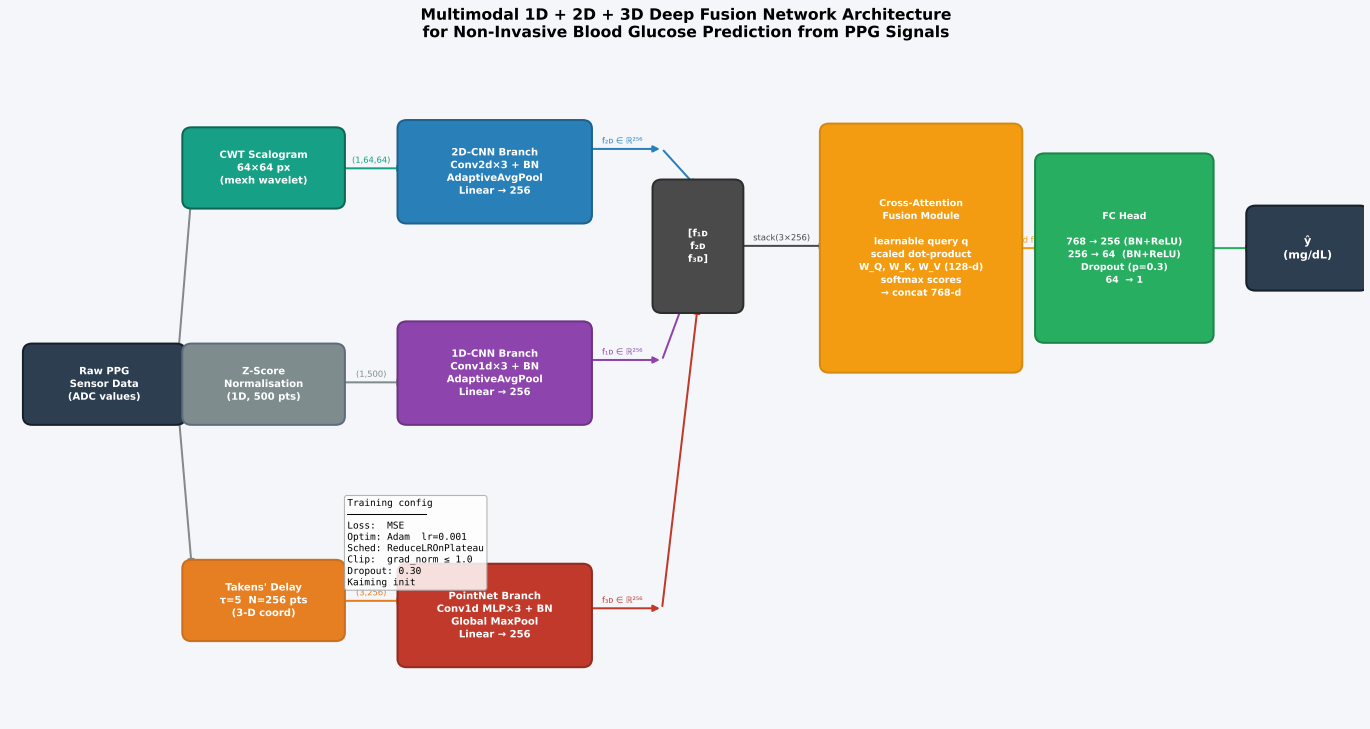


Fig. 1: Proposed Multimodal 1D+2D+3D Fusion Network architecture. The raw PPG signal is simultaneously transformed into three heterogeneous representations feeding equal-dimension (256-d) branch encoders. A cross-attention module with a learnable query vector fuses the branches into a 768-d representation passed to the FC regression head.

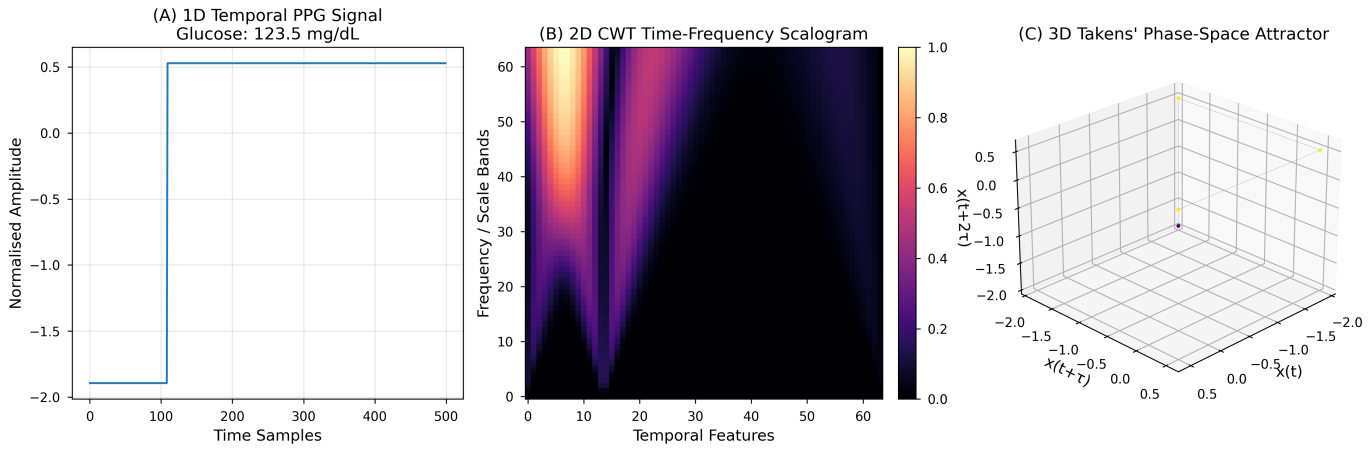


Fig. 2: Example multimodal transformation of a single PPG window (sample index 350, glucose ≈115 mg/dL). **(A)** Z-score normalised 1D temporal signal; **(B)** 2D CWT time-frequency scalogram (64×64 pixels, Mexican hat wavelet); **(C)** 3D Takens delay-embedding phase-space attractor ($\tau = 5$, $N_p = 256$ points), coloured by the third coordinate.

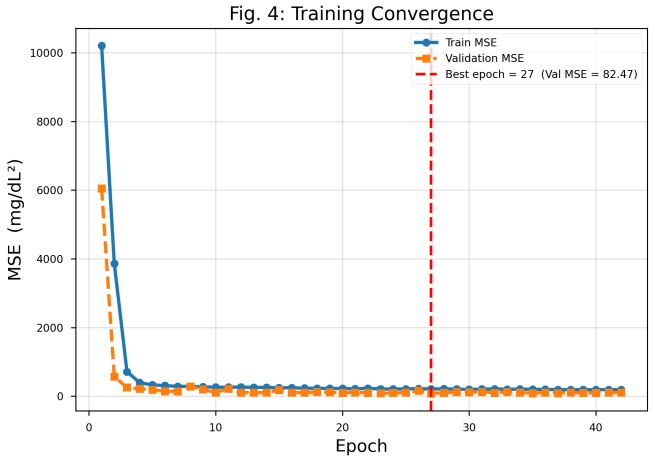


Fig. 3: Training convergence over 42 epochs (early stopped). ReduceLROnPlateau halved the LR at epochs 19 and 33 (vertical arrows not shown). Best validation MSE of 82.47 mg/dL² achieved at epoch 27.

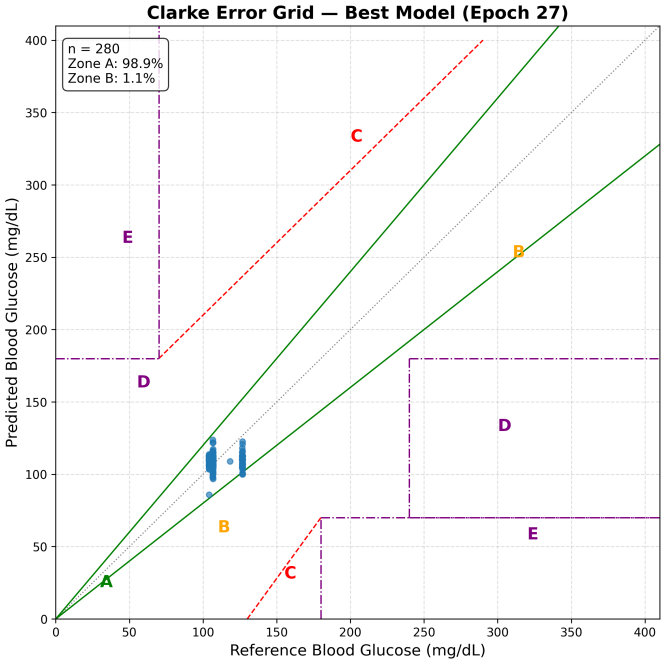


Fig. 4: Clarke Error Grid for 280 validation predictions. 277 points (98.9%) lie in Zone A; 3 points (1.1%) lie in Zone B; zero points in Zones C–E, indicating strong estimation agreement under the evaluated conditions.

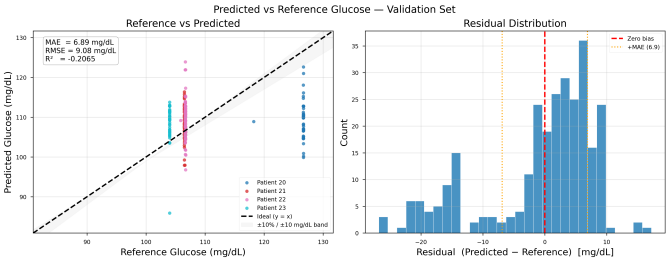


Fig. 5: Predicted vs. reference glucose scatter (left) and residual histogram (right) for the 280-window validation set. Points are colour-coded by patient (patients 19–22). MAE = 6.89 mg/dL, RMSE = 9.08 mg/dL. The residual distribution is approximately zero-centred, indicating minimal systematic bias.

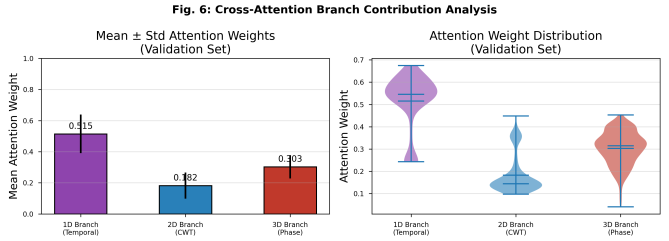


Fig. 6: Cross-attention branch weight distribution across 280 validation windows. Left: mean ± std bar chart; Right: violin plots. The 1D temporal branch receives the highest mean weight (0.515 ± 0.124), followed by the 3D phase-space branch (0.303 ± 0.075) and the 2D CWT branch (0.182 ± 0.084).

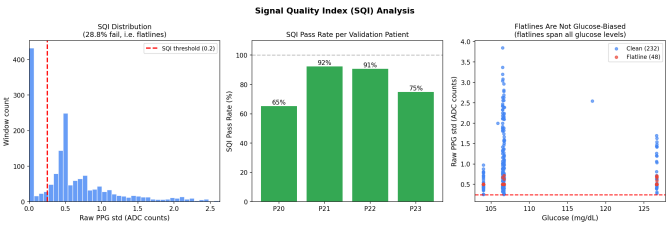


Fig. 7: Signal Quality Index analysis. **Left**: distribution of per-window raw PPG standard deviation across all windows (threshold $\tau = 0.25$, dashed red). **Centre**: per-patient SQI pass rates (patients 19–22). **Right**: SQI vs. glucose scatter; flatlines are not glucose-biased, confirming artefact-random occurrence.

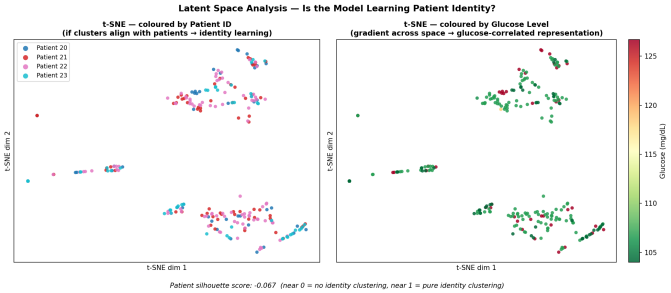


Fig. 8: t-SNE of the 768-d fused latent space ($n = 280$ val windows). **Left**: coloured by patient ID; silhouette score -0.067 confirms no patient-identity clustering. **Right**: coloured by glucose label, showing a diffuse glucose gradient rather than sharp identity-driven boundaries.

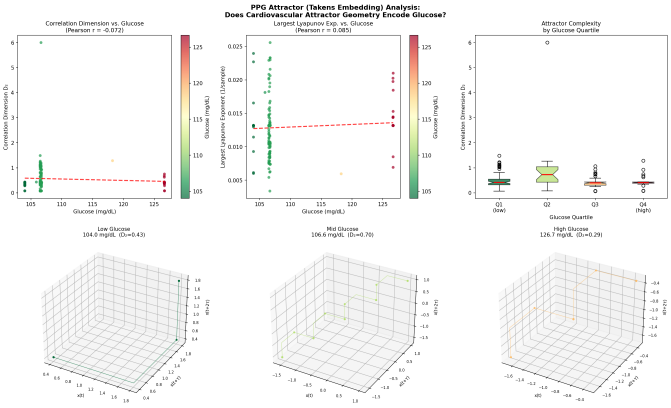


Fig. 9: Cardiovascular attractor analysis for SQI-clean validation windows. **Top row**: correlation dimension D_2 and largest Lyapunov exponent (LLE) versus glucose (Pearson $r = -0.072$ / $r = +0.085$). **Bottom row**: example 3D Takens attractors for low, mid, and high glucose windows.

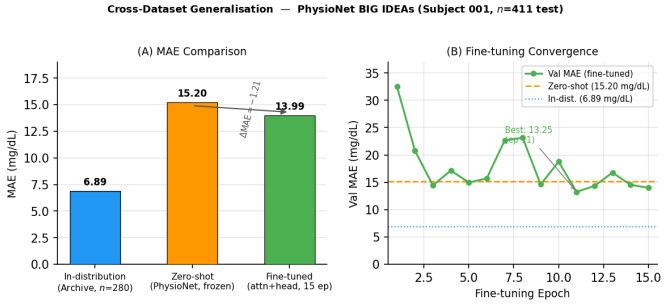


Fig. 10: Cross-dataset generalisation experiment (PhysioNet BIG IDEAs, Subject 001, Empatica E4 BVP 64 Hz). **(A)** MAE comparison: in-distribution Archive validation (6.89 mg/dL, blue); zero-shot transfer with all weights frozen (15.20 mg/dL, orange); fine-tuned cross-attention and head only (13.99 mg/dL, green). The ΔMAE arrow shows -1.21 mg/dL improvement from fine-tuning. **(B)** Per-epoch validation MAE during 15-epoch fine-tuning (Adam, η=10^{-3}, batch 32). Best val MAE 13.25 mg/dL at epoch 11; final reported MAE 13.99 mg/dL at epoch 15. Orange dashed: zero-shot baseline (15.20 mg/dL).